

Types of Monotonicity for Operators

Monotone

$$\langle x - y | u - v \rangle \geq 0$$

Paramonotone

Both A and A^{-1} monotone

Strictly Monotone

Monotone + inequality strict when $x \neq y$

Uniformly Monotone

$$\langle x - y | u - v \rangle \geq \phi(\|x - y\|)$$

Strongly Monotone

$$\langle x - y | u - v \rangle \geq \beta \|x - y\|^2$$

Cyclically Monotone

$$\sum_{i=1}^n \langle x_{i+1} - x_i | u_i \rangle \leq 0$$

Relationships:

Strongly Monotone $\Rightarrow$ Uniformly Monotone $\Rightarrow$ Monotone

Subdifferential of Convex Function $\Rightarrow$ Cyclically Monotone